

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE: ARTIFICIAL INTELLIGENCE

COURSE CODE: CSA17

CONSTRAINT-BASED PROBLEM SOLVING

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Course Outcome	CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively
Assessment	Assessment Tool 2 – Annexure A

Note: The uploaded question paper gives the constraints for Questions 1–3. For Questions 2 and 3, the exact grid/obstacle coordinates are not printed in the supplied image. Therefore, the answer explains the complete algorithm and gives a clearly marked representative grid execution where exact coordinates are required.

QUESTION 1 – HOSPITAL DOCTOR SHIFT ASSIGNMENT

Problem: A hospital has to assign three doctors (D1, D2, D3) to three shifts: Morning (M), Afternoon (A), and Night (N). Exactly one doctor must be assigned to each shift.

1.1 Constraints

- D1 cannot work the Night shift. Therefore, $D1 \in \{M, A\}$.
- D2 must work before D3: $\text{shift}(D2) < \text{shift}(D3)$, where $M < A < N$.
- Only one doctor can work each shift.
- D3 cannot work the Morning shift. Therefore, $D3 \in \{A, N\}$.
- All three doctors must be assigned exactly one shift.

1.2 CSP Representation

Variable	Domain	Restriction
D1	{M, A}	$D1 \neq N$
D2	{M, A, N}	D2 occurs before D3
D3	{A, N}	$D3 \neq M$

The **all-different** constraint is also applied: D1, D2 and D3 must receive different shifts.

1.3 Backtracking / Constraint Checking

We assign a shift to one doctor at a time. After every assignment, the constraints are checked. A branch is rejected immediately if it violates any constraint.

Step	Assignment	Check	Result
1	D1 = M	D1 is not Night	Valid
2	D3 = A	D3 is not Morning	Valid
3	D2 = N	D2 must be before D3	Reject: N is after A
4	D2 = M	D2 is before D3, but M is already used	Reject: duplicate shift
5	D1 = A	Try D3 = N	Valid so far
6	D2 = M	$M < N$ and all shifts are different	Valid solution

1.4 Complete Valid Schedules

Solution	Morning	Afternoon	Night	Constraint Check
A	D2	D1	D3	D1 not N; D3 not M; D2 before D3; all different
B	D2	D3	—	Incomplete if D1 must also be assigned

Final valid schedule: Morning = D2, Afternoon = D1, Night = D3.

A second valid schedule also exists: Morning = D1, Afternoon = D2, Night = D3. It satisfies $D1 \neq N$, $D3 \neq M$, and D2 works before D3. Hence the problem has **two valid complete schedules**.

1.5 Final Answer

Recommended schedule: D1 → Afternoon, D2 → Morning, D3 → Night. This satisfies every stated constraint and assigns exactly one doctor to each shift.

QUESTION 2 – ROBOT GRID NAVIGATION USING BFS

Problem: A robot operates in a 5×5 grid and must move from Start (S) to Goal (G). Obstacles block some paths.

2.1 Given Constraints

- The robot can move only Up, Down, Left and Right. Diagonal movement is not allowed.
- Cost of each move = 1.
- Blocked cells cannot be entered.
- Manhattan distance is used as a heuristic/informative measure: $|r_1 - r_2| + |c_1 - c_2|$.

2.2 Search Method: Breadth-First Search

BFS is suitable because every legal movement has the same cost (1). BFS expands nodes level by level. Therefore, the first time the goal is reached, the path contains the minimum possible number of moves.

Important: BFS uses a FIFO queue rather than a priority queue. If a priority queue is required, that is more closely associated with Uniform Cost Search. Since every edge has cost 1, BFS and Uniform Cost Search produce the same optimal path length.

2.3 Representative 5×5 Grid

The uploaded question does not show the exact obstacle coordinates. The following grid is used only to demonstrate the required BFS calculation.

Row\Col	1	2	3	4	5
1	S	.	#	.	.
2	#	.	#	.	#
3	.	.	.	.	.
4	.	#	#	#	.
5	.	.	.	.	G

= blocked cell; . = free cell.

2.4 BFS Step-by-Step Expansion

Level	Nodes expanded/discovered at that level	Path cost g(n)
0	(1,1) = S	0
1	(1,2)	1
2	(2,2)	2
3	(3,2)	3
4	(3,1), (3,3)	4
5	(4,1), (3,4)	5
6	(5,1), (2,4), (3,5)	6
7	(5,2), (1,4), (4,5)	7
8	(5,3), (1,5), (5,5)=G	8

BFS stops when G is removed from the queue/reached. Parent pointers are then followed backwards from G to S.

2.5 Optimal Path

$S \rightarrow (1,2) \rightarrow (2,2) \rightarrow (3,2) \rightarrow (3,3) \rightarrow (3,4) \rightarrow (3,5) \rightarrow (4,5) \rightarrow G$

Number of moves = 8. Since every move costs 1, total path cost = $8 \times 1 = 8$.

2.6 Manhattan Distance

For a node (r,c) and goal G=(5,5), Manhattan distance is $h(n) = |r-5| + |c-5|$. For example, $h(1,1)=8$ and $h(3,4)=3$. This can indicate how far a node is from the goal, but standard BFS does not use $h(n)$ to decide which node is expanded

next.

2.7 Final Answer

BFS is optimal here because all movement costs are equal. The representative execution reaches G with a minimum cost of **8**. If the actual assessment sheet contains a different hidden/printed obstacle layout, the same BFS procedure must be applied to that exact grid.

QUESTION 3 – AUTONOMOUS RESCUE ROBOT IN A PARTIALLY OBSERVABLE BUILDING

Problem: An autonomous rescue robot operates in a disaster-affected building. Some rooms are blocked, and the robot must reach a survivor located at G. The robot has limited sensing and must minimize total cost while navigating safely.

3(a) CONSTRAINT ANALYSIS

Constraint	Effect on decision-making
Movement only Up, Down, Left, Right	The robot cannot move diagonally. At every state it generates only four possible movement actions, subject to boundary constraints.
Blocked cells cannot be traversed	Blocked cells are removed from the legal action set. A path entering an obstacle is immediately rejected.
Limited sensing: only adjacent cells are observed	The robot cannot know the complete map initially. It must make decisions using local observations and update its internal map as it moves.
Every movement costs 1	Shorter routes are preferred because each extra move increases total cost.
Entering risky/uncertain areas has additional cost	The robot must consider both distance and risk. A slightly longer safe route may be cheaper than a short risky route.
Minimize total cost while ensuring safety	The objective is not simply minimum distance; it is minimum accumulated movement + risk cost subject to safety constraints.

The problem is therefore a **cost-sensitive path-planning problem under partial observability**. The robot must repeatedly sense, update its knowledge, choose an action, observe the result, and re-plan when necessary.

3(b) SOLUTION APPROACH – UNIFORM COST SEARCH / ONLINE SEARCH

A suitable approach is **Uniform Cost Search (UCS)** when the relevant map and edge costs can be represented in the search state. UCS always expands the lowest accumulated-cost node first, so it is optimal when all step costs are non-negative.

For a transition $n \rightarrow n'$, the path cost can be written as:

$$g(n') = g(n) + \text{movement cost} + \text{risk cost}$$

Here, movement cost = 1. If the new cell is risky/uncertain, risk cost = 2. Therefore a normal move costs 1 and a risky move costs 3.

Because sensing is limited, an **online version** is appropriate: the robot does not assume that it knows every obstacle in advance. It plans from the current knowledge, moves, senses adjacent cells, updates its internal map, and replans if the newly discovered information changes the best route.

3(b).1 Algorithm Steps

1. Initialize the robot at S and set accumulated cost $g(S)=0$.
2. Sense the four adjacent cells: Up, Down, Left and Right.
3. Mark observed blocked cells as unavailable and observed free/risky cells with their corresponding costs.
4. Insert legal neighboring states into a cost-priority queue.
5. Remove the state with the smallest accumulated cost.
6. If the selected state is G, stop; the corresponding parent chain gives the minimum-cost known route.
7. Otherwise, sense again because new information may have become available.
8. If a newly discovered obstacle invalidates the planned route, update the map and re-run UCS from the current position.
9. Continue until the survivor at G is reached.

3(b).2 Illustrative Cost Comparison

Route	Moves	Risky cells	Cost calculation	Total
Route A	8	0	$8 \times 1 + 0 \times 2$	8
Route B	6	2	$6 \times 1 + 2 \times 2$	10
Route C	9	1	$9 \times 1 + 1 \times 2$	11

In this example, Route A is selected because its total cost is 8, even though another route may contain fewer physical moves. The important point is that the robot minimizes **total cost**, not distance alone.

3(c) APPLICATION OF AI CONCEPTS

AI Concept	Application in the rescue robot
Intelligent agent	The robot perceives the environment through sensors and chooses actions that help it reach the survivor safely.
Rationality	The robot selects the action expected to minimize total cost while satisfying safety constraints.
Environment type	The environment is partially observable, dynamic/uncertain due to disaster conditions, sequential, and potentially stochastic.
State	Current robot position plus the robot's current knowledge about observed free, blocked and risky cells.
Actions	Move Up, Down, Left or Right; sensing is part of the perception process.
Goal test	The robot reaches the survivor's location G.
Path cost	Sum of movement costs and additional risk costs along the route.
Search / planning	UCS or an online replanning strategy is used to select a low-cost safe route.

3(d) JUSTIFICATION OF THE CHOSEN APPROACH

- UCS directly handles different path costs. A risky cell can cost more than an ordinary cell, so BFS is not sufficient when costs are unequal.
- UCS is optimal for non-negative step costs because it expands the least-cost frontier first.
- Online replanning is important because the robot observes only adjacent cells and therefore cannot rely on a completely known map.
- The approach respects blocked-cell constraints and prevents the robot from intentionally selecting an unavailable path.
- The method balances movement efficiency and safety by incorporating risk penalties into the path cost.

Therefore, **online Uniform Cost Search with continuous sensing and replanning** is a suitable solution. It is more appropriate than simple BFS because the problem contains unequal costs and partial observability.

3(e) FINAL CONCLUSION

The rescue robot should behave as a rational intelligent agent: sense the nearby environment, maintain an internal map, generate only legal actions, assign movement and risk costs, select the lowest-cost safe route using UCS, and re-plan whenever new observations reveal an obstacle or risk. This provides a systematic and safe way to reach the survivor.

QUICK REVISION / KEYWORDS TO WRITE IN THE EXAM

Question	Keywords / Core Points
Q1 – CSP	Variables = D1,D2,D3; domains = M,A,N; constraints = D1≠N, D3≠M, D2<D3, all-different; solve by backtracking; valid schedule
Q2 – BFS	Equal step cost; FIFO queue; expand level by level; ignore blocked cells; 4-direction movement; first goal reached gives shortest path
Q3 – Rescue robot	Partial observability; local sensing; blocked cells; movement cost 1; risky-cell penalty 2; minimize total cost; intelligent rational agent

STUDENT DETAILS

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Submission note: Questions 2 and 3 in the supplied image do not contain a visible exact obstacle/grid layout. The worked grid in Q2 and numerical cost comparison in Q3 are therefore explicitly presented as illustrative demonstrations. The algorithmic answers and constraint analysis remain applicable to the actual grid supplied by the faculty.